KINGSMEAD 33/11 kV PRIMARY SUBSTATION AND 11 kV NETWORK - DATA SHEET (sheet 1 of 4)

Arrangement (single-line diagram, sheet 3). The 33 kV grid infeed supplies the 33 kV busbar P33. Transformers T1 and T2 (33/11 kV) feed 11 kV busbar sections A and B, which are joined by the bus-section circuit breaker BS. Feeders F1 and F2 leave section A, feeders F3 and F4 leave section B. Secondary substations S1 to S12 are 11 kV nodes. Interconnectors IC-1 (S3 to S10) and IC-2 (S6 to S11) and the normally-open points NOP-1 (S4 to S8) and NOP-2 (S7 to S12) link the feeders. Every route, its cable sections, joints and lengths are on the cable route plan (sheet 4); each cable section has one cable type throughout.

33 kV grid infeed: a source of 33.6 kV (line) behind an impedance of 0.36 + j3.60 ohm per phase. The infeed is metered at the 33 kV busbar P33: power drawn from the infeed means the power flowing from the grid into P33.

Transformers T1, T2: each 15 MVA, 33/11 kV, impedance 10.0 % on 15 MVA with X/R = 20, referred to the 11 kV side and the same on every tap. On-load tap changer on the 33 kV winding, taps 1 to 17, tap 9 nominal; each tap step changes the 33 kV winding turns by 1.25 % of nominal; a higher tap number means more 33 kV turns (lower 11 kV voltage). The ratio at tap n is 33 (1 + 0.0125 (n - 9)) : 11 kV. Magnetising current and core loss are negligible.

Automatic voltage control (AVC). A transformer whose AVC is in AUTO runs on the tap that brings the voltage of the 11 kV busbar section it feeds closest to 11.20 kV, with every other tap position fixed as it is. In MANUAL the tap stays where the operator left it. When BS is closed both sections are one busbar.

Cables (11 kV, XLPE, aluminium, three-core), per phase at 90 C:
  type 300: R 0.130 ohm/km, X 0.093 ohm/km, rating 465 A
  type 185: R 0.211 ohm/km, X 0.097 ohm/km, rating 360 A
  type 95: R 0.411 ohm/km, X 0.108 ohm/km, rating 245 A
  type 70: R 0.568 ohm/km, X 0.112 ohm/km, rating 205 A
Cable shunt capacitance, switch and joint impedances are negligible. The loading of a cable section is its current as a percentage of its type rating. A closed route is a series impedance equal to the sum of its sections.

Loads at 09:00 (constant power, lagging power factor):
  S1 1450 kW pf 0.95 | S2 980 kW pf 0.93 | S3 1210 kW pf 0.94 | S4 760 kW pf 0.92
  S5 1630 kW pf 0.95 | S6 1120 kW pf 0.90 | S7 690 kW pf 0.91 | S8 1380 kW pf 0.95
  S9 1040 kW pf 0.93 | S10 850 kW pf 0.92 | S11 1720 kW pf 0.96 | S12 910 kW pf 0.93

Solar farm PV-1 at S10: available output at 09:00 is 4000 kW; it exports the lower of the available output and any export limit in force, at a fixed power factor of 0.95 absorbing reactive power (it draws reactive power from the network).

Capacitor bank C1 at S6: 1200 kvar at 11.0 kV, a fixed shunt capacitance (its kvar varies with the square of the S6 voltage).

Notes.
1. Report steady-state values for the network as the log leaves it at 09:00.
2. Switching state changes only through operations recorded in the log.
3. Voltages are line-to-line; currents are phase currents; power flows are three-phase.
4. Network losses = 33 kV infeed active power (metered at P33) + PV-1 export - total load.

KINGSMEAD 11 kV CONTROL LOG - NIGHT SHIFT 21/22 NOV (sheet 2 of 4)

22:00 Handover. T1 and T2 in service, both AVC AUTO. BS closed. IC-1 and IC-2 closed. NOP-1 and NOP-2 open. C1 out. PV-1 not generating.

22:30 Permit 4471: T2 tap-changer diverter inspection. T2 33 kV and 11 kV CBs opened. BS remains closed; T1 carries both sections.

23:15 Rounds. All feeder currents normal.

01:20 F1 trip and reclose. Contractor damage reported on F1-3 between joint J7 and S3. F1-3 isolated: switches opened at S2 and S3. NOP-1 closed to back-feed S4 from S8.

02:05 Repair crew on site F1-3 (permit 4475).

03:40 T1 AVC selected MANUAL for tap-changer counter check; T1 set to tap 9 and left in MANUAL.

05:10 C1 at S6 switched in for voltage support.

06:30 Permit 4471 cancelled. T2 33 kV CB closed, T2 11 kV CB closed onto section B. BS opened for T2 return-to-service checks. T2 AVC selected AUTO. T1 AVC remains MANUAL.

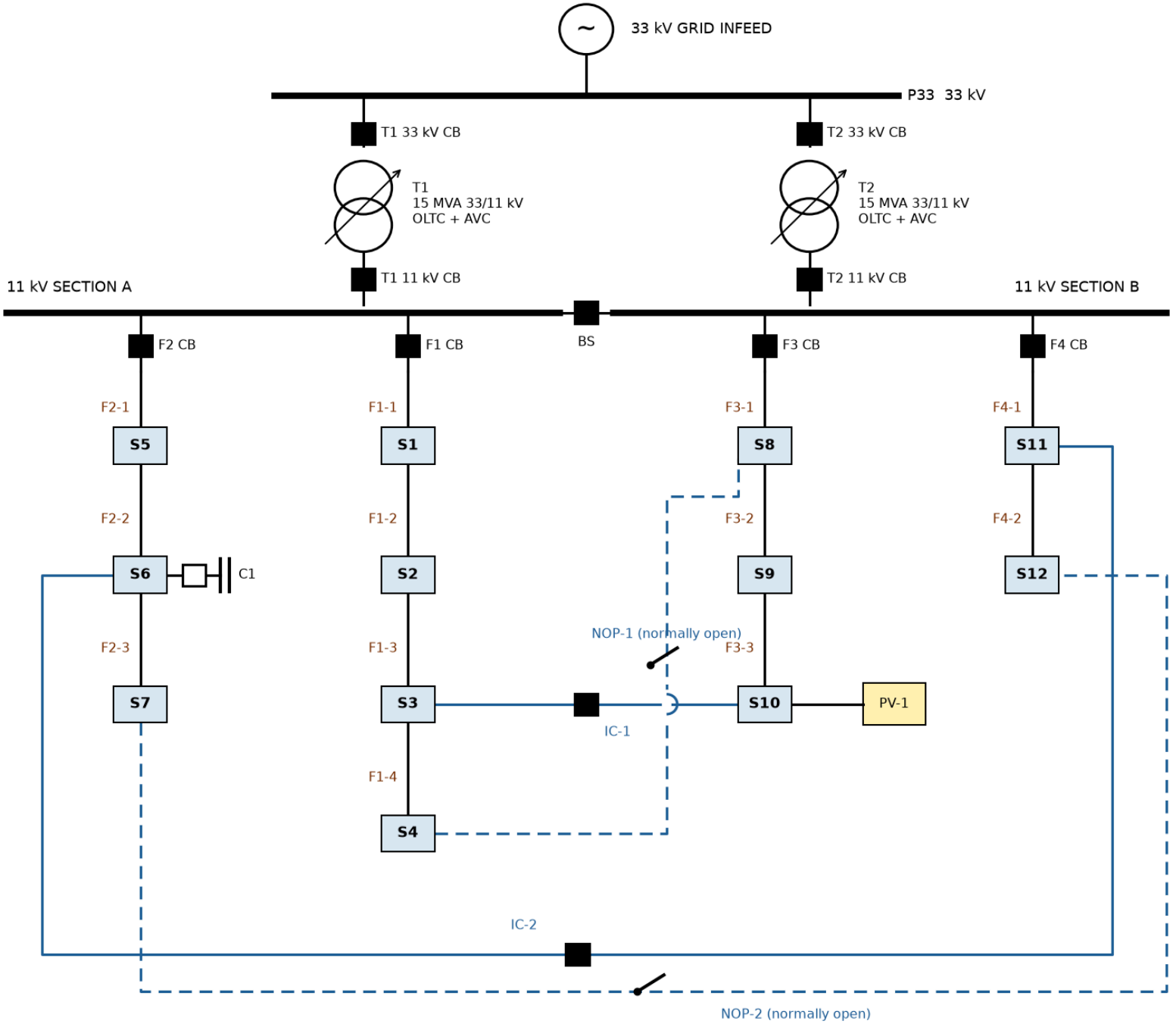
07:20 Permit 4475 cancelled. F1-3 repaired: the damaged 95 mm2 section between J7 and S3 (380 m) cut out and replaced by 410 m of 185 mm2 Al laid in a new trench. F1-3 tested and switches at S2 and S3 closed. NOP-1 left closed per control engineer (ring running closed).

07:50 Planned transfer of S7 to F4: NOP-2 closed. Opening of F2-3 at S6 postponed (switching operator recalled to another site); F2-3 remains closed.

08:15 DNO instruction: PV-1 export limited to 2500 kW until further notice. Power factor setting unchanged.

08:40 Rounds. No alarms. No further switching.

09:00 Handover to day shift.



KINGSMEAD 11 kV - sheet 3: single-line diagram (schematic, not to scale). Breakers and switches shown in their NORMAL state:
filled square = closed, open square = open, open blade = normally-open point. Route ids (F1-1 …) refer to the cable route plan, sheet 4.

KINGSMEAD 11 kV - sheet 4: cable route plan (NOT to scale; lengths as labelled). Each route is shown section by section with cable type and section length; joints are bars with joint numbers. Brown: feeder routes; blue: interconnectors; dashed: normally-open routes.

